KUWAIT'S NEWBORN SCREENINGNATIONAL PROGRAM REVIEW (2021)

ABSTRACT

Newborn Screening is a Public Health Program refers to special tests done to infants 24 – 48 hours after birth, aiming to Screening, Diagnosis, Management, Evaluation and Education of serious and life threatening Genetic, Endocrine and Metabolic health disorders that don't show signs at birth.The diseases included in the screening program do not manifest at birth and have a window period of a few days to months, this give a chance to screen for these conditions and to initiate treatment at the pre-clinical stage to obtain best outcome. If these babies left untreated, this leading to potentially fatal or disabling condition including mental retardation, motor impairment or physical disabilities. Early diagnosis and treatment also prevent anxiety for parents as well as costs associated with seeking diagnosis for unexplained symptoms. Most treatments are inexpensive and may involve the addition of a vitamin to the diet, Hormone supplementation, avoidance of certain foods and chemicals or dietary changes. Newborn Screening or Newborn Blood Spot Screening (NBS) involves taking few drops of blood from the heel of the newborn on a special card. This card will then sent to the laboratory to be tested for special biochemical markers of special disorders. Results then evaluated according to the protocol of Newborn Screening Program for each country.

Historical background

The origins of newborn screening date back to the mid-20th century with the work of Dr. Robert Guthrie, who developed a simple, inexpensive and effective test to determine whether newborns had phenylketonuria (PKU). In order to define the requirements that need to be met to include additional diseases in screening programmes, the World Health Organization published the Wilson and Jungner criteria for screening within Principles and practice of screening for disease in 1968.[1] These criteria, defined according to current circumstances, aimed to guarantee that these programs fulfil their primary objective: maximum benefit with minimum cost (table1).

As time passed, new laboratory tests were developed for detection of additional diseases, among which we ought to emphasise congenital hypothyroidism (CHT). Phenylketonuria and congenital hypothyroidism are the diseases most commonly screened for, and there is substantial variation in the remaining diseases that are included in screening panels. After a few years in which NBS programmes advanced slowly, as additional diseases were included at the pace that technological advances occurred, there was a revolution in these programmes in the early 1990s with the introduction in screening laboratories of tandem mass spectrometry (MS/MS).[2]

(Table 1): Wilson and Jungner Classic Screening Criteria.[3]

1. The condition should be an important health problem.
2. An accepted treatment with a recognized disease exists.
3. Facilities for diagnosis and treatment should be available.
4. There should be a recognizable latent or early symptomatic stage.
5. There should be a suitable test or examination.
6. The test should be acceptable to the populations.
7. The natural history of the condition (at all stages) should be adequately understood.
8. There should be an agreed policy on whom to treat.
9. The cost of case-finding (including diagnosis and treatment) should be economically balanced in relation to possible expenditure on medical care as a whole.
10. Case-finding should be an on-going process and not a "once and for all" project.

Analytical Method

Tandem MS (or MS/MS),Itis a powerful analytical technique used in the newborn screaning due to its sensitivity, specifity and ability to measure multiple analyte in one analytical run. LC-MS/MSis a non dervatised chromsystem which allows simple, fast and semi-quantitative determination of amino acids, acylcarnitines and succinylacetonein dried blood spot (DBS) through simple sample preparation and efficient extraction of the analyte without additional butylation step.

The Principle of Tandem mass spectrometry is based on coupling mass spectrometers together in a series to analyze complex mixtures. The method uses two mass filters arranged sequentially with a collision cell between them. The filters can be used in static or scanning mode to select a particular mass-to charge (m/z) ratio or m/z range.[4]The analyte first have to be ionized (Electrospray Ionization "ESI")to generate a mixture of ions and then transferred into gas phase.

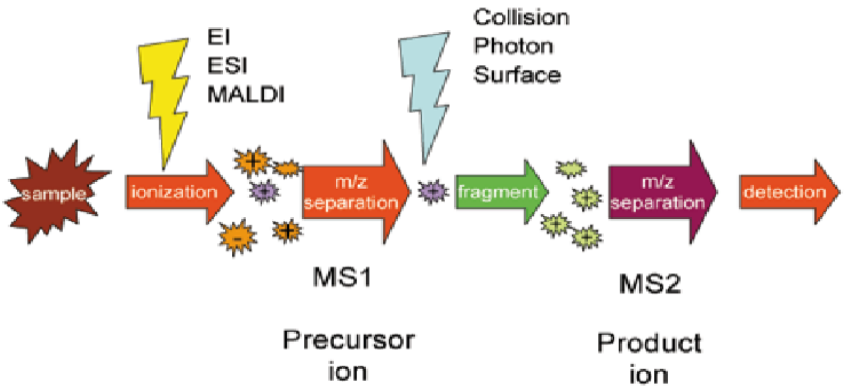
The triple quadruple Tandem mass spectrometer uses a variety of measurement modes the one presented here is (Multiple Reaction Monitoring "MRM) mode. In MRM mode both first and second mass filters are set statically to a particular mass-charge ratio. The molecules of a given sample are ionized and thefirst spectrometer MS1 selects the ions by their mass to charge ratio (Ions with different mass-charge ratio are not shown) .In the collision cell, ions of aparticular m/z ratio coming from MS1 are selected and then made to split into smaller fragment ions. These fragments are then introduced into the second mass spectrometer (MS2), which in turn separates the fragments by their m/z ratio and detects them[5](Fig. 1).

Because of its high selectivity, the application does not require a HPLC column, allowing very short run times (2 min).

To compensate interfering ion suppression effect and to allow precise quantification of the analyte the system uses isotopically labeled standard.

Because of the prescence of variation like type of the filter paper and blood volume on the punched out dried blood spot (sample hematocrit), it is a semi-quantitative method and so it should not be used as the only laboratory diagnostic method and should followed by confirmation diagnostic methods based on molecular-genetic, enzymatic or based on chromatography.

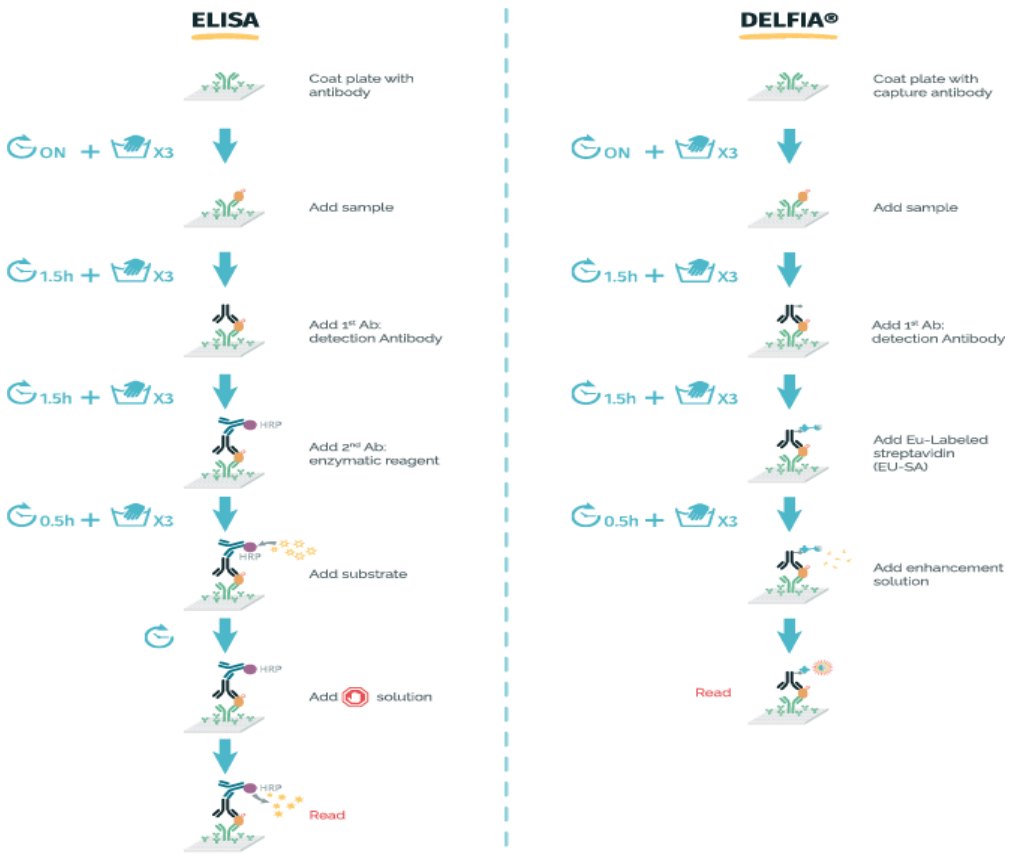
Figure 1: Explains how Tandem MS works.[6]



DELFIA® (dissociation-enhanced lanthanide fluorescence immunoassay),

DELFIA is a proven, robust immunodetection platform based on time-resolved fluorescence (TRF) detection. This detection method, coupled with the unique chemical properties of lanthanide chelates, makes DELFIA a superior alternative to conventional ELISA for research, drug discovery and clinical applications. DELFIA offers a combination of high sensitivity, wide dynamic range and long-lasting stability and excellent Flexibility that traditional ELISA cannot match (Fig. 2).

Figure 2: Workflow comparison between a DELFIA TRF assays and atraditional ELISA.[7]



DELFIA system is used for sensitive and precise quantification of four assays to diagnose four diseases. These diseases include: Congenital hypothyroidism (hTSH), Congenital adrenal hyperplasia (17-α-OHP), Galactosemia (GAO, GALT) and Biotinidase deficiency (BTD).

Neonatal hTSHassay: It is used for quantitative determination of human thyroid stimulating hormone in a dried blood sample. It is based on a solid phase , two site fluoroimmunmetric assay based on the direct sandwich technique in which two monoclonal antibodies, one immobilized and one europium-labeled monoclonal antibodies, in the assay buffer are directed against two antigenic determinants on hTSH molecule. An assay buffer elutes hTSH from dried blood spot. After incubation and wash, Enhancement solution is added which dissociates europium ions from the labelled antibody into the solution where they form a highly fluorescent chelates with component of the enhancement solution. The fluorescence of each sample is then measured and it is **directly proportional** to concentration of hTSH in the sample. The complete assay requires only one incubation step.[8,9]

17-α-OH-Progesterone assay (17-α-OHP): It is used for quantitative determination of human 17-α-OH-progesterone in a dried blood sample. It Is based on a solid phase, time resolved fluoroimmunassay based on the competition between europium- labeled 17-α-OHP and sample 17-α-OHP for a limited number of binding sites on 17-α-OHP specific polyclonal antibodies. A second antibody, directed against the IgG is coated to the solid phase, leading to separation of antibody bound and free antigen. After incubation and washing, Enhancement solution is added which dissociates europium ions from labeled antigen into the solution where they form highly fluorescent chelates with the component of the solution. The fluorescence in each well is measured which is inversely proportional to 17-α-OHP concentration in sample.[8,9]

Biotinidase assay (BTD): It is based on a fluorometric assay used for semi-quantitative determination of biotinidase activity in the dried blood specimen. Biotinidase enzyme in the blood sample utilizes biotin 6-aminoquinoline (BAQ) as substrate. BAQ is cleaved to fluorescent product 6-aminoquinoline (6-AQ) and it is measured with a plate fluorometer. The enzyme activity is given in enzyme units 1U= (1 nmol /min/dl).[10]

Total Galactose assay (GAO): It is used for quantitative determination of total galactose (galactose and galactose 1-phosphate) in the dried blood spot. It is based on the use of a fluorescent galactose oxidase method. It is a series of reaction ending in the production of 3-(p-hydroxyphenol) propionic acid. The fluorescence can be measured using excitation central wavelenghthof 405 nm.[11]

Galactose-1-phosphate uridyltransferase (GALT): It is used for the semi-quantitative determination of neonatalgalactose -1- phosphate uridyltransferase enzyme concentration in the dried blood specimen. It is an adaptation of the semi quantitative enzymatic assay of Beutler and Baluda[12] which results in production of flurescent NADPH. After preciptation with ethanol, the fluorescence is measured using an excitation wavelength of 355 nm and an emission wavelength of 460 nm.

DNA-based Techniques, DNA techniques have also been introduced to newborn screening programs in recent years. . DNA arrays and sequencing have proved a more accurate tool for newborn screening programs, however, they also can add to the cost of screening.[13]

Dried Blood Spot Collection and Residual Retention

Within 24–72 hours of birth, virtually every baby has a heel stick performed, which begins the newborn screening process. This heel stick is taken by a health care professional and blood is spotted onto a piece of filter-paper card and dried before being sent to the newborn screening laboratory for testing.[14]When the sample is received, The blood spots is tested and demographic information from the newborn screening card entered into a database . This database links the infants information with the results of the screening tests. These results are reviewed by a biochemistry consultant to determine if the infant has a lower risk of having the disease (screen negative) or a higher risk of having the disease (screen positive).

Newborn screening: Experiences in the Middle East and North Africa (MENA) region:

Consanguinity and endogamy are high in the Middle East-North Africa (MENA) region including the Gulf countries and range between 20% and 60%. In many MENA countries, consanguineous marriages are culturally favored with longstanding traditions. Together with consanguinity, the relatively large family size plays a role in the high prevalence of autosomal recessive conditions in the population of the region. Additionally, the high rate of consanguinity favors the coexistence of multiple distinct autosomal recessive disorders within single families. Policy makers should endorse social, educational, and public health initiatives to mitigate the impact of genetic disease in the MENA region, as well as establishing preventive programs like premarital and newborn screening. For this reason four meeting had held, first three MENA meetings in Marrakech (2006), Cairo (2008), and Doha (2010), organised by and with substantial financial support from the US National Institute for Child Health and Human Development (NICHD), the fourth Meeting was held in March 8–11, 2020 by online meeting due to covid-19 pandemic. This meeting provides a common forum for all of the regional screening programs to find information and resources, and for an open exchange of experience and expertise that benefits programs at an early stage of development, as well as those that are already at an advanced stage.[15]

NDBS programs function within a comprehensive system that includes education, screening, short-term follow-up, diagnosis, treatment/management, and long-term follow-up/evaluation.[16] The NBS system is often challenged by economic, political, and cultural considerations. The initiation of NBS in developing health systems, such as many in the Middle East and North Africa (MENA) and the Asia Pacific (AP) regions, has been slow for various reasons, including lack of understanding by the individual as well as family, societal, and financial benefits. All countries with NBS either have faced or will face challenges inimplementing NBS; however, many developing health systems face additional challenges related to the economy, government stability, culture/religion, geography, and health/political priorities. [17]

The following (TEN) challenges to successfully implement a sustainable NBS program: (1)Planning, (2)Leadership,(3)Education, (4)Medical support, (5)Technical support, (6)Logistical support,(7)Protocol/policy development,(8) Administration,(9)Evaluation ,(10) and sustainability.

These ten challenges have been successfully dealt with to varying degrees in all developed and developing NDBS programs.[18]

Current state of Newborn Screening in the Gulf Countries

SAUDI ARABIA:

In 1994 metabolic screening lab in King Faisal Specialist Hospital & Research Centre (KFSHRC) introduced expanded newborn screening for amino acid disorders, fatty acid oxidation and organic acidemia. In 2005 the Saudi newborn screening program start. In 2016, the government of Saudi Arabia launched the National Transformation Program to ensure the realisation of Kingdom's Vision 2030. According to the Saudi Ministry of Health, the program aims to screen all newborns within 24–72 h of delivery, and is currently performed in 183 hospitals all over the kingdom.[19]Upon its establishment, 16 inborn errors of metabolism and endocrine disorders are screened for which are summarized in (Table 2). In 2016, screening for hearing-loss and critical congenital heart defects was added to newborn screening, which is currently performed in 30 referral Ministry of Health hospitals.

The workflow of the Saudi newborn screening program, including procedures concerning repeat of screening tests for initially positive results, confirmation using biochemical testing and referral system to specialised hospitals for management plan initiation..[20]

(Table 2): List of inborn errors of metabolism and endocrine disorders screened for within the National Newborn Screening Program in Saudi Arabia.[21]

Aminoacidopathies	Phenylketonuria (PKU)
	Argininosuccinatelyase deficiency (ASL)
	Maple syrup urine disease (MSUD)
	Citrullinaemia
Organic acid disorders	Propionic acidaemia (PPA),
	Methylmalonicacidaemia,
	Glutaricacidaemia type-I
	Isovalericacidaemia
	3-methylcrotonyl-CoA carboxylase deficiency (3-MCC)
Fatty acid oxidation defect	Medium-chain acyl CoA dehydrogenase deficiency
Ketogenesis and ketolysis defects	3-hydroxy-3-methylglutaryl-CoA lyase deficiency
	Beta-ketothiolase deficiency (BKT)
Carbohydrate disorder	Galactosaemia (GALT)
Endocrine disorders	Congenital hypothyroidism (CH)
	Congenital adrenal hyperplasia (CAH)
Vitamin responsive disorder	Biotinidase deficiency.

QATAR

In Qatar the rate of consanguinity is relatively high with a rate of 54%, and predominantly first cousin marriages comprising 26.7% of all marriages.The Metabolic Laboratory at Hamad Medical Corporation serves as the National Newborn Screening Laboratory for the State of Qatar. Approximately 28,000 newborns per year are screened in Qatar. Screening is provided for all newborns of residents at no cost. Typically more than 98% of the births are screened. The current screening panel of 83 disorders performed by the Laboratory includes Galactosemia, Biotinidase Deficiency, Congenital Hypothyroidism, Congenital Adrenal Hyperplasia, 25 Amino Acid Disorders, 16 Fatty Acid Oxidation Disorders,

25 Organic Acidemias, and 12 Hemoglobinopathies. GuanidinoacetateMethyltransferase Deficiency, Severe Combined Immunodeficiency, and 4 Lysosomal Storage Diseases (LSD)are currently under validation. The screens are performed using a wide range of technologies including photometric techniques, tandem mass spectrometry, and liquid chromatography. 2nd tier testing that provides methylmalonic acid and methylcitric acid levels are performed as a reflex for suspicious screens with abnormal levels of methionine, homocysteine, or propionylcarnitine to differentiate a disorder and to reduce false positives. The Laboratory is currently evaluating and validating a reflex to a rapid targeted next generation sequencing to improve performance metrics.

The Metabolic Laboratory in Qatar is currently validating a digital microfluidic system for screening for Pompe, Fabry, Gaucher, and Mucopolysaccharidosis Type I.False positive rates associated with LSD screening are a significant concern. The Metabolic Laboratory has established a screening protocol that incorporates targeted next generation sequencing (tNGS) as a reflex confirmatory test for all suspected positive screens for LSD. From the newborn screening laboratory experience in Qatar, advances in molecular technology and the expansion of the recommended uniform newborn screening panel of diseases have led to earlier life-saving treatment and intervention.[22]

United Arab Emirates (UAE):

The National Neonatal Screening Program in United Arab Emirates was launched in January 1995, through the Central Department of Maternal and Child Health and covered the following diseases at the national level: Phenylketonuria (1995), Congenital hypothyroidism (1998), Sickle cell disease (2002), Congenital Adrenal Hyperplasia (2005), Biotinidase Deficiency (2010), Tandem mass spectrometry (MS/MS) for AA, FA&OA (2011), Galactosemia (2016) & G6PD deficiency (2019). Plus Cystic Fibrosis as pilot study in 2020.

Currently, every newborn is screened for more than 40 diseases including congenital hypothyroidism, congenital adrenal hyperplasia, biotinidase deficiency, phenylketunurea, sickle cell disease, β Thalassemia diseases, Galactosemia, Glucose-6-Phosphate Dehydrogenase Deficiency, plus group ofAmino Acids, Organic Acid and Fatty Acid disorders. This is in addition to congenital heart disease and hearing screening programs and Cystic Fibrosis as pilot study.

The protocol includes systematic outcome evaluation of all phases (Pre-analytical, analytical and post-analytical). The quality assessment of the National Neonatal Screening Laboratory is under the Centers for Diseases Control and Prevention (CDC) evaluation.[23]

Oman

Like other Gulf countries, Oman has 52% consanguinity andmetabolic diseases are at least 3 to 5 times more common. Birthrate in the country is about 45,000 per annum, which will requireat least 123 screening tests per day.Oman has a good healthcare network of 57 hospitals and 136health care centers. The National Congenital Hypothyroidism screeninghas been in operation since 2004. Tandem MS has been in operationat the Sultan Qaboos University Hospital for metabolic diagnosticservices since 2002 giving the institution a wealth of clinical andtechnical expertise.Main barriers to screening program in Oman are cost, availabilityof diagnostic and treatment facilities, public and healthcare workereducation, difficulties in reaching many geographical locations, andrecalling positive cases. However,thefull commitment of the Ministry of Health to bear the cost ofscreening and disease management is necessary.[24]

From the experience of running metabolic disease services in Oman, a program is proposed incorporating common diseases that can be treated by simple and cost effectivemeans. After a period of 3 to 5 years' experience, more diseases can be added gradually to reach an expanded program.

A. Initial program

TMS tests: Phenylketonuria, IsovalericAcidemia, Type 1 GlutaricAciduria, Medium Chain Acyl CoA Dehydrogenase (MCAD) deficiency, HMG CoA lyase deficiency.

AutoDELFIA tests: Congenital Hypothyroidism, Galactosemia, Biotinidase deficiency, Congenital Adrenal Hyperplasia.

Hematological diseases: Sickle Cell Disease, thalassaemia, G6PD deficiency.

B. Expanded program

TMS tests:Phenylketonuria, Homocystinura, Maple Syrup UrineDisease, Arginino Succinic Aciduria, Citrullinemia, PropionicAcidemia, Methyl MalonicAcidemia, IsovalericAcidemia, BetaKetothiolase deficiency, GlutaricAciduria types 1 & 2, MediumChain Acyl-CoA Dehydrogenase deficiency, Very Long Chain AcylCoA Dehydrogenase deficiency, HMG-CoA Lyase deficiency.

AutoDELFIA tests:Galactosemia, Biotinidase deficiency, Cysticfibrosis, Hypothyroidism, Congenital Adrenal Hyperplasia.

Hematological diseases: Sickle Cell Disease, Thalassaemia, G6PDdeficiency.[25]

Newborn Screening Program In Kuwait

Kuwait is a country located in the Northern edge of Eastern Arabia at the tip of the Arabian Gulf between Iraq and Saudi Arabia with total area of 17.818 Km^2. As of 2022, Kuwait has a population of 4.67 million people of which 1.45 million are Kuwaiti citizens while the remaining are foreign nationals from over 100 countries. Kuwait is a developing country with a high income economy.[26,27]

Kuwait has one of the highest consanguinity rates in the world with arate of 54% . Social, religious, cultural, political and economic factors still play roles in favouring consanguineous marriages. Several research studies in Kuwait have shown that marriages among relatives is one of the major factors leading to health problems because of increase the autosomal recessive disorders.[28]

The idea of newborn screening in Kuwait was born in 1979 when Kuwait Medical Genetic Centre (KMGC) has constructed with screening for phenylketonuria as a pilot studies then increased in 2005 to include screening for Phenylketonuria (PKU) and Congenital hypothyroidism (CH) after receiving DELFIA system. In 2007 it includes 5 disorders PKU, CH, 17 Hydroxy progesterone (17OHP), Galactosemia and Biotinidase Deficiency. In October 2014, Expanded newborn screening Program began for screening of a wide range of Metabolic and Endocrine Disorders, including 22 disorders via testing dried blood spot (DBS), after receiving Tandem Mass spectrometry system making Kuwait has one of the most comprehensive newborn screening programs in the Middle East (Table 3,4).

Note that, Newborn Screening Program (NBS) in Kuwait does not include screening for Thalassemia and Glucose 6 Phosphate Dehydrogenase Deficiency (G6PD).

Thalassemia is a group of hereditary disorders resulting from defect in the rate of synthesis of one or more globin chain (α or β chain) leading to decrease normal HB (HB A) and so increase (ɣ chain) production which means increase Fetal Haemoglobin (Hb F) as a compensation. (Hb F) is the main Hb from 9 months till birth. At birth it forms 70-90% of total Hb.

Switch from Hb F to Hb A starts 3 – 6 months after birth, because of that it does not included in our program which performed during the **FIRST MONTH OF LIFE**.

G6PD is an X-Linked recessive disorder resulting from deficiency of enzyme glucose 6 phosphate dehydrogenase which can lead to certain type of anaemia called Hemolytic anaemia.

G6PD is not included in the screening program because it is already done in Governmental Hospital as a routine if there is a **FAMILY HISTORY**.

(Table.3): Kuwait Newborn Screening program development and expantion.

Year	No. of filter paper received for screening	No. of Disorders Screened
2005	3029	2 Disorders Screened as Pilot Study After receiving DELFIA system: • Phenylketonurea • Congenital Hypothyroidism
2006	2547	
2007	2438	5 Disorders Screened as Pilot Study After receiving DELFIA system: • Phenylketonuria • Congenital Hypothyroidism • Congenital Adrenal hyperplasia • Classic galactosemia • Biotinidase deficiency
2008	4714	
2009	15287	
2010	17692	
2011	25228	
2012	29779	
2013	29517	
2014	31987	Expanded program of 22 disorders started in October 2014 After receiving TANDEM system.
2015	52789	
2016	57951	
2017	59655	
2018	55210	
2019	56333	
2020	56441	
2021	57600	

(Table 4): List of endocrine and metabolic disorders included in the National Newborn Screening Program in Kuwait indicating primary and secondary marker for each.

Group	Disorder	Primary markers	Secondary markers
Endocrine Disorders (2 disorders)	Congenital hypothyroidism (CH)	**hTSH**	_
	Congenital adrenal hyperplasia (CAH)	**17a-OH-P**	_
Amino Acid Disorders (6 disorders)	Maple Syrup Urine Disease (MSUD)	**LEUCINE**	**VALINE**
	Phenylketonurea (PKU)	**PHENYLALANINE**	**PHE/TYROSINE**
	Citrullinemia (CIT-1)	**CITRULLINE**	_
	Tyrosinemia Type 1 (TYR-I)	**TYROSINE**	_
	Classical Homocystinuria (CHCY)	**METHIONINE**	_
	ArgininosuccinincAcidemia (ASA)	**METHIONINE**	_
Fatty Acid Oxidation Disorders (5 Disorders)	Long-Chain 3-Hydroxy Acyl-CoA Dehydrogenase Deficiency(LCHAD)	**C16OH**	**C18OH C16OH/C16**
	Medium-Chain Acyl-CoA Dehydrogenase Deficiency (MCAD)	**C8**	**C8/C2 C8/C10 C10:1**

	Trifunctional Protein Deficiency (TFP)	**C16OH**	**C18OH** **C16OH/C16**
	Very Long Chain Acyl-CoA Dehydrogenase Deficiency (VLCAD)	**C14:1**	**C14** **C14:1/C2** **C14:1/C16** **C14:2**
	3-Hydroxy-3-methylglutaryl-CoA Lyase Deficiency (3HMG)	**C5OH**	**_**
Organic Acid Disorders (8 disorders)	Glutaric acidemia type I (GA-I)	**C5DC**	**_**
	Isovaleric acidemia (IVA)	**C5**	**C5/C2** **C5/C3**
	Methylmalonic acidemia (MMA)	**C3**	**C3/C2**
	Multiple CoA Carboxylase Deficiency (MCD)	**C5OH**	**_**
	Beta ketothiolase deficiency (BKT)	**C5OH**	**_**
	Propionic Acidemia (PA)	**C3**	**C3/C2**
	3-Methylcrotonyl-CoA Carboxylase deficiency(3MCC)	**C5OH**	**_**
Galactosemia	Classic galactosemia	**GALT** **GALT.**	**_**
Biotinidase Deficiency		**BIOTINIDASE Enzyme**	**_**

Newborn Screening Program (NBS) in Kuwait initially only covered 5 Public Hospitals until October 2014 when the private hospitals also included except Royal Hayat and Al-Rashid Hospital. In June 2019, Royal Hayat Hospital also included in the program and Kuwait Medical Genetic Center (KMGC) become the center of NBS program, at NBS Laboratory that covering 5 public Hospitals and 14 private Hospitals.

Newborn screening program in Kuwait cover 100% of newborns inside Kuwait, receiving samples from all Governmental and Private hospitals. (Table 5) showing list of 5 Governmental and 14 Private Hospitals in Kuwait.

(Table 5): Governmental and private hospitals in Kuwait

Governmental Hospitals	Al-Jahra Hospital	Al-Adan Hospital
	Al-Farwanyia Hospital	Maternity Hospital
		Jaber Hosptal

Private Hospitals	Al-Salam Hospital	Royal Hayat Hospital
	Alia Hospital	Al-Orf Hospital
	Al-Mowasat Hospital	International Hospital
	Tiba Hospital	Al-Hady Hospital
	Al-Seif Hospital	Dar-Elshifa Hospital
	Sidra Hospital	London Hospital
	KOC Hospital	Kuwait Hospital

While all of the diseases tested for are rare and not usually apparent at birth, a relatively large number of affected infants in Kuwait will be found to have these conditions every year. Screening program can help these children to have the best start in life through timely newborn screening, early diagnosis and treatment. If the infant is screen positive, this does NOT mean that

the infant has a disease; however, it does mean that the infant has an increased chance to have a disease.

TESTS PERFORMED WITH TANDEM MASS SPECTROPHOTOMETRY

Amino Acid disorders

These metabolic disorders are caused by body's inability to breakdown certain amino acids in proteins, or by the inability to detoxify the by-product of amino acids (ammonia) through the urea cycle due to deficiency or absence of the concerned enzymes. The buildup of amino acids and/or by-products of amino acid metabolism in the blood cause severe brain damage or even death.

Amino Acid disorders may be detected using this panel:

- Maple syrup urine disease (MSUD) (Ketoacid dehydrogenase deficiency).
- Phenylketonuria (PKU) (Phenylalanine hydroxylase deficiency).
- Citrullinemia (CIT-1) (Argininosuccinic acid synthetasedeficiency).
- Tyrosinemia Type I (TYR 1) (Fumarylacetoacetate hydrolase deficiency).
- Classical Homocystinuria (CHCY) (Cystathionine B synthase deficiency).
- Argininosuccinic academia (ASA) (Argininosuccinyl-COA lyase deficiency).

Fatty acid oxidation disorders

Are a group of inherited metabolic conditions occur when fatty acids cannot be completely metabolized to produce energy this leading to accumulation of fatty acids, and a decrease in cell energy metabolism. Each fatty acid oxidation disorder is associated with a specific enzyme defect in the fatty acid metabolic pathway and affects utilization of dietary and stored fat.

fatty acid oxidation disorders can be detected using this panel.

- Long Chain 3-Hydroxy Acyl-CoA Dehydrogenase Deficiency (LCHAD) (Long chain L-3-hydroxyacyl-COA dehydrogenase deficiency).
- Medium Chain Acyl-CoA Dehydrogenase Deficiency (MCAD) (Medium Chain Acyl-CoA Dehydrogenase deficiency).
- Trifunctional Protein Deficiency (TFP) (Long Chain Acyl-COA dedeficiency).
- Very Long Chain Acyl-CoA Dehydrogenase Deficiency (VLCAD) (Very Long Chain Acyl-CoA Dehydrogenase Deficiency).
- 3-Hydroxy-3-methylglutaryl-COA Lyase Deficiency (3HMG) (3-Hydroxy-3-methylglutaryl-COA Lyase deficiency.

Organic acid disorders (Acylcarnitine)

Are a group of inherited metabolic conditions.associated with a specific enzymesinvolved in breakdown of amino acids deficient or absent. This causes the accumulation of organic acids in blood and urine. The accumulated compounds or their metabolites are toxic, resulting in the clinical features of these disorders.

organic acid disorders may be detected using this panel:

- Glutaricacidemia type I (GA-1) (Glutaryl-COA dehydrogenase deficiency).
- Isovalericacademia (IVA)(Isovaleryl-COA dehydrogenase deficiency (IVA).
- Methylmalonicacademia (MMA) (Methylmalonyl-COA mutase deficiency).
- Multiple COA carboxylase deficiency (MCD) (Holocarboxylasesynthetase deficiency).
- Beta-ketothiolase deficiency (BKT) (B-Ketothiolase deficiency)
- Propionic acidemia (PA) (propionyl-COA carboxylase)
- 3-methylcrotonyl-CoA carboxylase deficiency (3MCC) (3-Methylcrotonyl-COA carboxylase deficiency).[29]

TESTS DONE WITHIN DELFIA LAB

Congenital Hypothyroidism (CH): It is defined as thyroid hormone deficiency present at birth. It must be diagnosed promptly by newborn screening because delay in treatment can lead to irreversible neurological deficits .The majority of CH (80%) caused by thyroid dysgenesis which is always sporadic or non hereditary. It can be also due to primary causes of thyroid or central origin. Treatment with thyroxine helps prevent mental and growth retardation.[30]

Congenital Adrenal Hyperplasia (CAH): Congenital adrenal hyperplasia (CAH) is an autosmal recessive inherited disorder of the adrenal cortex occur due to deficiency of 21-

hydroxylase enzyme characterized by cortisol deficiency, with or without aldosterone deficiency, and androgen excess. Patients with the most severe form also have abnormalities of the adrenal medulla and epinephrine deficiency.[31]Symptoms of classic CAH include ambiguous genitalia in females, both male and female can suffer of loss of appetite, nausea, fatigue and it may leads to emergency "adrenal crisis"which can cause low blood pressure and blood sugar,dehydration, vomiting and diarrhea

Galactosemia: Galactosemia is a rare autosomal recessive inherited genetic disorder of carbohydrate metabolism that affects the body's ability to convert galactose to glucose**.** The most common and clinically severe form is classic glactosemia which caused by a deficiency of an enzyme galactose-1-phosphate uridyl-ltransferase (GALT) which is vital to this process. Classic galactosemia and clinical variant galactosemia can both result in life-threatening health problems including vomiting excessively, enlargement of the liver (hepatomegaly), appearance of amino acids and protein in the urine, growth failure, accumulation of fluid in the abdominal cavity (ascites) with abdominal swelling (edema) may also occur. Diarrhea, irritability, lethargy and a bacterial infection may also be early signs of galactosemia.

Early diagnosis and treatment with a lactose-restricted (dairy-free) diet is absolutely essential to avoid profound intellectual disability, liver failure and death in the newborn period. A diet low in galactose can prevent irreversible brain damage and other complications.[32]

Biotinidase deficiency (BTD): Biotinidase deficiency (BIOT) is an autosomal recessive inherited condition in which the body is unable to use and recycle the vitamin biotin. Because the body needs free biotin as a cofactor for carboxylase enzymes to break down fats, proteins, and carbohydrates effectively.

There are two main types of biotinidase deficiency (BTD), differing in the severity of signs: severe "profound biotinidase deficiency" and mild "partial biotinidase deficiency."
Signs of BTD usually start within a few months after birth. In some cases, the symptoms may not appear until childhood. This is why early screening and identification is so important.
Early signs of BTD include: Seizures, Weak muscle tone (hypotonia), Trouble breathing, Skin rash,Hair loss, Trouble balancing, A fungal infection called candidiasis.Many of these signs can be triggered by illnesses or infections. Treatment with biotin will be lifesaving.[33]

Cutoff values, Quality Assurance and Quality Control Program:

Internal quality control run and evaluated with each batch of tests.
Also the laboratory participate in the Newborn Screening Quality Assurance Program (NSQAP) run by the Centers of Disease Control and Prevention (CDC), USA. (NSQAP) at the Centers for Disease Control and Prevention (CDC) is an accredited proficiency testing (PT) provider (ISO/IEC 17043:2010).

NSQAP provides comprehensive dried blood spot (DBS) PT materials every 3 months **(QUARTERY)** used by laboratories to achieve certification by regulatory bodies, and quality control materials (QC) every 6 months **(BIANNUAL)** for monitoring method performance.

Tandem biomarkers cutoff values of the tests calculated depnding on validation of cut off from Newborn CDC proficiency testing (PT) that sent regularly to our Lab . This started in 2017 because of missed cases resulting from the previous cut off values (Table 6).

(Table 6): Cutoff values for each biomarker done by TANDEM LAB

Analyte	1ry& 2ry Markers	Our Cutoff	Unit	Disorder
LEU	1ry	254	µM/L	Maple Syrup Urine Disease (MSUD)
VAL	2ry	180	µM/L	
PHE	1ry	123	µM/L	Phenylketonurea (PKU)
PHE/TYR	2ry	2.2	µM/L	
CIT	1ry	43	µM/L	Citrullinemia (CIT-1)
TYR	1ry	265	µM/L	Tyrosinemia Type 1(TYR-1)
MET	1ry	44	µM/L	Classical Homocystinurea (CHCY)
C3	1ry	6.4	µM/L	Methylmalnic Acidemia(MMA) Propionic Acidemia (PA)
C3/C2	2ry	0.28	µM/L	
C5	1ry	0.73	µM/L	Isovaleric Acidemia (IVA)
C5/C2	2ry	0.04	µM/L	
C5/C3	2ry	0.44	µM/L	
C5OH	1ry	0.95	µM/L	Multiple Carboxylase Deficiency (MCD)
C5DC	1ry	0.52	µM/L	Glutaric academia type 1(GA-1)
C8	1ry	0.13	µM/L	Medium-Chain Acyl-CoA Dehydrogenase Deficiency (MCAD)
C8/C2	2ry	0.01	µM/L	
C8/C10	2ry	2.01	µM/L	
C10:1	2ry	0.11	µM/L	
C16OH	1ry	0.07	µM/L	Long-Chain-acyl-COA dehydrogenase deficiency Trifunctional protein deficiency
C18OH	2ry	0.06	µM/L	
C16OH/C16	2ry	0.07		
C14:1	1ry	0.29	µM/L	Very Long Chain Acyl-CoA Dehydrogenase Deficiency (VLCAD)
C14	2ry	0.47	µM/L µM/L	
C14:1/C2	2ry	0.03	µM/L	
C14:1C16	2ry	0.22	µM/L	
C14:2	2ry	0.2	µM/L	

For DELFIA biomarkers (GALT, T.GAL, TSH, 17OHP and BIOTINIDASE), cutoff values: It is from the kit inserts.

Note that, for17-α-OH-P cut off values are modified to be weight and age related and for TSH cut off values are age related (Tables 7, 8, 9) **.**

(Table 7): Cut off values for 17-α-OH-P(Age & Wight Related)

Disorder	Weight (Grams)	Age (days)	RR (nmol/l)
17-α-OHP	<1000	0-19	<200
		20-29	<100
		30-59	<60
		>60	<30
	1000-1500	0-3	<150
		4-13	<120
		14-19	<80
		20-29	<60
		30-59	<40
		>60	<30
	1500-2000	0-3	<80
		4-13	<60
		14-29	<40
		>30	<30
	2000-2500	0-1	<60
		2-3	<50
		4-13	<40
		>14	<30
	>2500	0-1	<60
		2-3	<40
		>4	<30

(Table 8): Cut off values for h-TSH (age related)

Disorder	Age (hrs)	Reference Range (µu/Ml)
h-TSH	0-23	<40
	24-48	<25
	>48	<15

(Table 9):Cut off values for GALT, T.Galactose, Biotinidase

Disorder	Low Risk Result	High Risk Result		Unit
	(Negative)	(Border Line)	(Positive)	
GALT	>3.5	2.5 – 3.5	<2.5	U/g TLB
T.GAL	<15	15-25	>25	Mg/dl
Biotindase	>60	16-60	<16	U

Sample collection and reporting of the results

Sample collected 1 day (24 hours) to 7days after birth, best time between 2days (48 hours) and 3 days (72 hours) after birth, at Newborn Screening Offices (NSO) located in the 5 Governmental hospitals and it is responsible for collection and transportation of samples, parent education, clinical assessment and doing confirmatory tests for screen positive cases, reporting

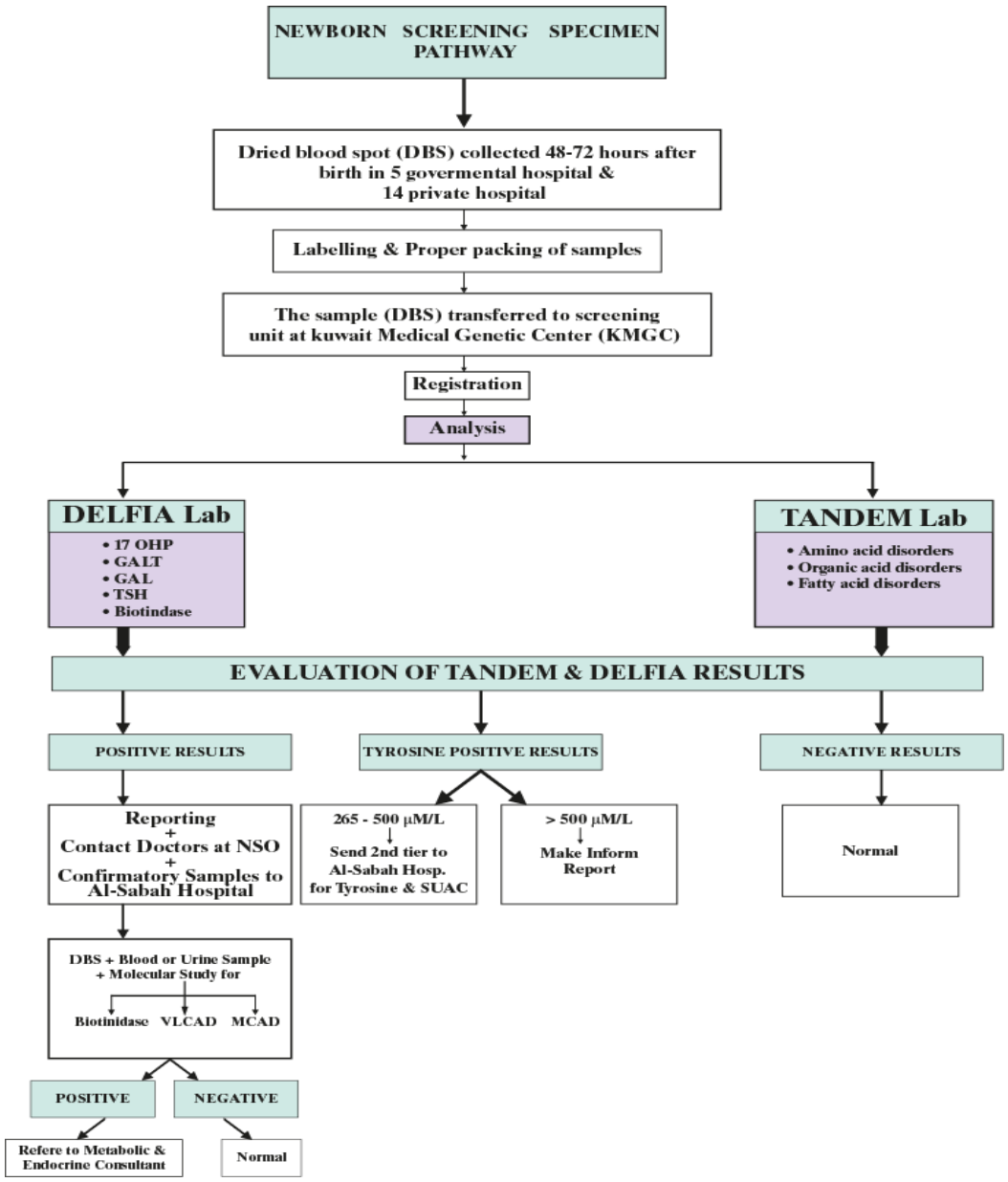
and referral of confirmed cases to specialists, quality assurance activity and monthly statistics to KMGC.

Special consideration

- Early collection, before 1day (24hours), the test repeated within 5 days.
- Premature baby(less than 33 weeks) collect three samples at 2, 14 and 30 days (premature may have false positive result).
- Newborns receive transfusion, some disorders may be missed, ideally collect before transfusion.
- Newborns receive (Total Parenteral Nutrition), amino acids and organic acids may be elevated
- Early discharge(before 24 hour) repeat screening should be done (The screen should be part of the discharge summary).

Reporting of the results 1 to 3 working days after receipt of specimen.False positive and false negative results are possible with this screening. Infants with presumptive positive screening tests require confirmatory tests and when confirmed, the clinician should immediately check on the clinical status of the baby and refer the infant to a metabolic disease specialist.(Fig. 3) showing pathway of specimen collection transportation, analysis and evaluation.

(Figure 3): Newborn screening specimen pathway



NB. Second tier test run on the **SAME SAMPLE** used for the primary screen, often a different methodology is used. It is done to identify infants at risk of having a metabolic condition, while reducing false positives and false negatives.[34]

In our Lab we make second tier test only for TYROSINE,

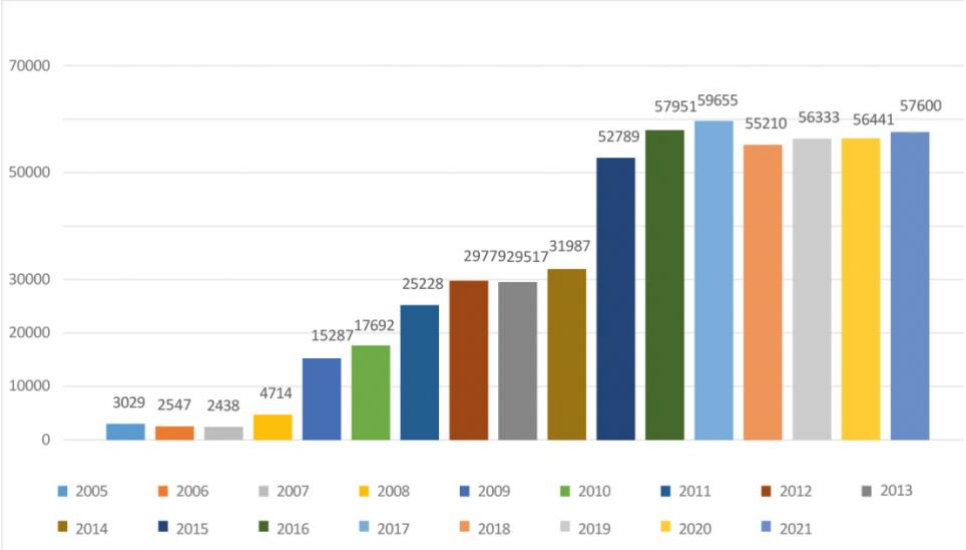
Tyrosine is a rare inherited disorder result from defect of the enzyme Fumarylacetoacetate Hydrolase necessary to break down Tyrosine resultIng in toxic accumulation of tyrosine metabolites fumarylacetoacetate and succinylacetone (SUAC).

According to our protocol, if the result of Tyrosine 265-500µM/L we send second tier sample to Al-Sabah Hospital for Tyrosine and Succinylacetone.

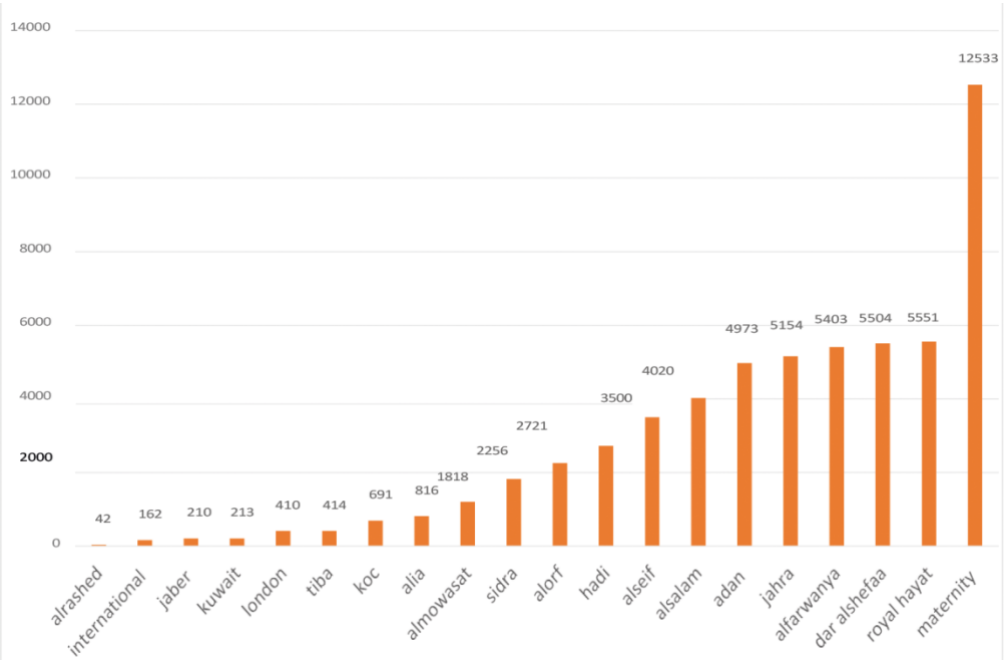
If the result more than 500µM/L we make an **inform report** (Fig.3)

Following Diagrams Show total number of Filter papers received for screening each year form 2005 – 2021. Sorting samples sources showing that maternity hospital is the highest in sending rate.

(Figure 4): Total number of filter papers received for screening each year from 2005 to 2021.



(Figure 5): Samples sources from lowest to highest showing that maternity hospital is the highest in sending rate (2021).



Turnaround time

The Time from receipt of sample to report by date of collection and date of birth. Some outliers in our turn-around-time reports are due to older children being screened as we receive samples collected from premature baby up to 30 days old.

Also there is transportation Time which is the difference between the date of collection (DOC) and the date in which the sample is received in the laboratory after drying the samples for four hours prior to sending, keeping in mind that the driver has a scheduled pick up once daily; So, any samples that are not yet dry will be delayed by at least 24 hours, also samples collected late in Thursday waiting for Saturday for transportation (Table 10).

(Table 10): Turnaround Time (2019 – 2021) "Available data"

Statistic	DOC to receive	Received to report	DOB to report	DOC to report
Average	1.09	2.38	7.6	3
Median	1	2	5	2.6
Mode	1	2	4	2.3

DOC: date of collection
DOB: date of birth

RESULTS:

From January 2015 to December 2021 the detection rate was: (1diseased newborn : from each 439 newborn). The following table shows the detection rate of each screened disorder in details, Positive predictive values for each marker from (2015-2021) after expansion also shown

(Table 11): Positive predictive values for markers from (2015-2021)

Disorder	positive screen	Confirmed postive	False negative	Detectionrate	FPR%	PPV%
CAH	1460	28	0	1:14,142	0.3%	1.92%
BIOT.Def	618	419	0	1:945	0.05%	67.8%
C.HYPOTHYRODISM	810	261	8	1:1,517	0.13%	32.2%
GAL	94	20	0	1:19,798	0.01%	21.3%
Citrullinemia+ASA	216	12	0	1:32,998	0.05%	5.5%
Tyrosiemia	792	10	0	1:39,597	0.19%	1.2%
Phenylketonuria	267	31	1	1:12,773	0.05%	11.6%
Homocystinuria/Hypermeth.	474	10	0	1:39,597	0.11%	2.1%
MSUD	199	7	0	1:56,568	0.04%	3.51%
MCAD	328	8	0	1:49,497	0.08%	2.43%
VLCAD	178	48	0	1:8249	0.03%	26.9%
LCHAD	23	1	0	1:395,979	0.00%	4.34%
GlutaricaciduriaI	67	5	0	1:79,195	0.02%	7.46%
IsovalericaciduriaIVA	192	3	0	1:131,993	0.04%	1.56%
PA+MMA+MCA	641	27	1	1:14,665	0.2%	4.21%
3HMG+BKT+MCC+MCD	147	10	0	1:39,597	0.03%	6.8%

(Total Confirmed Cases from 2015 To 2021) : 900 case

Summary

Newborn screening (NBS) is an essential public health strategy that enables the early detection and management of several congenital disorders, which if left untreated, may lead to mental retardation and/or death. Early diagnosis and initiation of treatment, along with appropriate long-term care help ensure normal growth and development and reduce costs to the healthcare system. Newborn screening began in 1960 when Robert Guthrie, developed a test for phenylketonuria (PKU), then the development of tandem mass spectrometry (MS/MS) facilitated the rapid expansion of newborn screening. MS/MS is a multi-analyte technique that allows numerous disorders to be detected from a single blood specimen more quickly, accurately, and cost-effectively than previous methods. Physicians can now diagnose more metabolic diseases before the onset of symptoms, thus preventing newborns from suffering death or irreversible adverse health conditions. Newborn Screening Positive results should be always undergo confirmatory tests based on Molecular-genetic or based on Chromatographic technique.

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